

MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.3: Eigenvalues, eigenvectors, and diagonalization

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Basic definitions

Definition Let A be an $n \times n$ matrix. Suppose $\mathbf{v}$ is a *nonzero* vector in $\mathbb{R}^n$ and λ is a scalar (which could be 0) such that

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Then λ is an **eigenvalue** of A , and $\mathbf{v}$ an **eigenvector** of A associated with the **eigenvalue** λ .

Activity

Check that

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

are eigenvectors for the matrix

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix},$$

and find the corresponding eigenvalues.

We plug in $\mathbf{v}_1$ into the matrix A , and calculate that

$$\begin{aligned} A\mathbf{v}_1 &= \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) + (-1)(1) \\ 6(1) + (-4)(1) \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2\mathbf{v}_1. \end{aligned}$$

Since $A\mathbf{v}_1 = 2\mathbf{v}_1$, the vector $\mathbf{v}_1$ is an eigenvector of A associated with the eigenvalue 2.

Next, we plug in $\mathbf{v}_2$, and check that

$$\begin{aligned}
 A\mathbf{v}_2 &= \begin{bmatrix} 3 & -1 \\ 6 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix} \\
 &= \begin{bmatrix} 3(1) + (-1)(6) \\ 6(1) + (-4)(6) \end{bmatrix} \\
 &= \begin{bmatrix} -3 \\ -18 \end{bmatrix} \\
 &= -3 \begin{bmatrix} 1 \\ 6 \end{bmatrix} = -3\mathbf{v}_2.
 \end{aligned}$$

Since $A\mathbf{v}_2 = -3\mathbf{v}_2$, $\mathbf{v}_2$ is an eigenvector of A associated with the eigenvalue -3 .

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}$$

be a matrix. Find the eigenvalues and eigenvectors of A .

We need to solve the equation $A\mathbf{v} = \lambda\mathbf{v}$, where both λ and $\mathbf{v} \neq \mathbf{0}$ are unknown.

The equation is equivalent to the equation $(\lambda I - A)\mathbf{v} = \mathbf{0}$. For each λ , in order for this equation to have a nontrivial solution $\mathbf{v}$, the matrix $\lambda I - A$ must be singular, i.e. $\det(\lambda I - A) = 0$. This allows us to temporarily eliminate $\mathbf{v}$ from the problem, determining the values of λ that have solutions $\mathbf{v}$.

We calculate that

$$\begin{aligned}\det(\lambda I - A) &= \det \begin{bmatrix} \lambda - 3 & 0 \\ 1 & 4 + \lambda \end{bmatrix} \\ &= (\lambda - 3)(\lambda + 4)\end{aligned}$$

Thus $\det(\lambda I - A) = 0$ holds precisely when $(\lambda - 3)(\lambda + 4) = 0$, i.e. when $\lambda = 3$ or $\lambda = -4$, and so these two values are the two eigenvalues of A . It remains to find the *eigenvectors* for each of the eigenvalues.

We now find the nontrivial solutions to the equation $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for $\lambda = 3$ and $\lambda = -4$.

Each of these is a linear system that can be solved, either using the methods of Anatomy of a Linear Map, or more ad hoc methods, which may be faster since the matrices we are dealing with are small.

For $\lambda = 3$, we calculate that

$$A - 3I = \begin{bmatrix} 0 & 0 \\ -1 & -7 \end{bmatrix}.$$

Thus the solutions to $(A - 3I)\mathbf{v} = 0$ are given by $-v_1 - 7v_2 = 0$, or $v_1 = -7v_2$. Thus the solutions to this equation are given by non-zero multiples of the basic solution

$$\begin{bmatrix} -7 \\ 1 \end{bmatrix}.$$

These are all eigenvectors of A with eigenvalue 3.

For $\lambda = -4$, we calculate that

$$A + 4I = \begin{bmatrix} 7 & 0 \\ -1 & 0 \end{bmatrix}.$$

Thus the solutions to $(A + 4I)\mathbf{v} = 0$ are those vectors $\mathbf{v}$ such that $v_1 = 0$, and so the eigenvectors for A with eigenvalue -4 are given by non-zero multiples of the basic solution

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

To conclude, the two eigenvalues of the matrix A are $\lambda = 3$ and $\lambda = -4$. The set of all eigenvectors of A associated with the eigenvalue $\lambda = 3$ is given by the expression

$$\left\{ \begin{bmatrix} -7t \\ t \end{bmatrix} : t \neq 0 \right\},$$

and the set of eigenvectors of A associated with the eigenvalue $\lambda = -4$ is given by

$$\left\{ \begin{bmatrix} 0 \\ t \end{bmatrix} : t \neq 0 \right\}.$$

Definition Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$\det(\lambda I_n - A) = \det \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix}$$

is called the

characteristic polynomial

of A and is sometimes

denoted $c_A(\lambda)$. The equation

$$\det(\lambda I_n - A) = 0$$

is called the

characteristic equation

of A .

Theorem Let A be an $n \times n$ matrix. The eigenvalues of A are the *roots* of the characteristic polynomial of A .

Remark Some authors define the characteristic polynomial to be $\det(A - \lambda I_n)$ rather than $\det(\lambda I_n - A)$. The advantage of the definition chosen in these notes is that the characteristic polynomial is always

monic

, i.e. the coefficient associated with the highest power of λ is always equal to one. The zeros of the two polynomials are both the same, and as we saw in Activity act-7-1-find-eigenvalues, they are the eigenvalues of A .

Activity

Find the characteristic polynomial of A if

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}.$$

We calculate that

$$\begin{aligned}\lambda I_3 - A &= \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda - 0 & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix} \\ &= \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix}\end{aligned}$$

The characteristic polynomial is therefore

$$\begin{aligned}
\det(\lambda I_3 - A) &= \det \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix} \\
&= (\lambda - 1) \det \begin{bmatrix} \lambda & -1 \\ 4 & \lambda - 5 \end{bmatrix} \\
&\quad - (-2) \det \begin{bmatrix} -1 & -1 \\ -4 & \lambda - 5 \end{bmatrix} \\
&\quad + (1) \det \begin{bmatrix} -1 & \lambda \\ -4 & 4 \end{bmatrix} \\
&= (\lambda - 1)[\lambda(\lambda - 5) - (-1)(4)] \\
&\quad - (-2)[(-1)(\lambda - 5) - (-1)(-4)] \\
&\quad + (1)[(-1)(4) - (\lambda)(-4)] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 4] \\
&\quad - (-2)[- \lambda + 1] + (1)[4\lambda - 4] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 4] \\
&\quad + (\lambda - 1)[-2] + (\lambda - 1)[4] \\
&= (\lambda - 1)[\lambda^2 - 5\lambda + 6] \\
&= \lambda^3 - 6\lambda^2 + 11\lambda - 6
\end{aligned}$$

Note: Finding eigenvalues and eigenvectors of a matrix A

1. Determine the roots of the characteristic polynomial $c_A(\lambda) = \det(\lambda I_n - A)$. These are the *eigenvalues* of A .
2. For each eigenvalue λ , find

all

 nontrivial solutions to the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$. These are the *eigenvectors* of A associated with λ .

Definition A basic solution (recall Basic solutions to homogeneous linear equations) to the homogeneous equation $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ will be called a

basic eigenvector

 associated with the eigenvalue λ .

Warning Do not make the common mistake of first row-reducing A to RREF and then finding the eigenvalues and eigenvectors of the RREF matrix. Row operations on the matrix A *do not* preserve the set of basic solutions. They only preserve solutions to the homogeneous equation $A\mathbf{v} = \mathbf{0}$.

If $\mathbf{v}$ is an eigenvector of a matrix A with eigenvalue λ , then any nonzero scalar multiple of $\mathbf{v}$ is also an eigenvector with the same eigenvalue λ . Indeed, if $\mathbf{w} = c\mathbf{v}$ for some scalar $c \neq 0$, then:

$$\begin{aligned} A\mathbf{w} &= A(c\mathbf{v}) \\ &= c(A\mathbf{v}) \quad (\text{by linearity of matrix multiplication}) \\ &= c(\lambda\mathbf{v}) \quad (\text{since } \mathbf{v} \text{ is an eigenvector with eigenvalue } \lambda) \\ &= \lambda(c\mathbf{v}) \\ &= \lambda\mathbf{w} \end{aligned}$$

Put differently, the entire line $\text{span}(\mathbf{v})$ is “invariant” under the map defined by A , with every vector in this line being stretched (or compressed) by the factor λ .

Fact Any non-zero linear combination of eigenvectors associated with the *same* eigenvalue λ is also an eigenvector with eigenvalue λ . Conversely, all eigenvectors associated with the eigenvalue λ can be written uniquely as a linear combinations of the basic eigenvectors associated with the eigenvalue λ .

Activity

Find the eigenvalues and basic eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

We begin by calculating the characteristic polynomial of A , i.e.

$$\det(\lambda I_3 - A) = \det \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{bmatrix}$$

Since this matrix is upper triangular, the determinant is the product of the diagonal entries, and so the characteristic polynomial is $c_A(\lambda) = (\lambda - 2)^3$. The polynomial has only a single zero, i.e. $\lambda = 2$, and so $\lambda = 2$ is the only eigenvalue of A .

To find the basic eigenvectors of A with eigenvalue 2, we solve the system $(2I - A)\mathbf{x} = \mathbf{0}$, i.e.

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus the solutions are given by $x_2 = 0$, with x_1 and x_3 as free variables. The set of basic eigenvectors is thus given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Diagonalization

Definition An $n \times n$ matrix A is called

diagonalizable

if there exists an invertible $n \times n$ matrix P

so that $P^{-1}AP$ is a diagonal matrix (recall Triangular and Diagonal Matrices). The matrix P is called a

diagonalizing matrix

for A .

Note that if $P^{-1}AP$ is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then for each standard basis vector (recall Standard basis) $P^{-1}AP\mathbf{e}_i = \lambda_i\mathbf{e}_i$. Multiplying this equation by P on the left hand side and setting $\mathbf{v}_i = P\mathbf{e}_i$ we obtain that

$$A\mathbf{v} = P(P^{-1}AP\mathbf{e}_i) = P(\lambda_i\mathbf{e}_i) = \lambda_i\mathbf{v}_i.$$

Thus $\mathbf{v}_i$ is an eigenvector of the matrix A associated with the eigenvalue λ_i . Conversely,

Activity

Suppose

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 0 \end{bmatrix}.$$

Task Compute $P^{-1}AP$ where

$$P = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix}.$$

You may use the fact that

$$P^{-1} = \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix}.$$

What does this computation say about the matrix A ?

We start by calculating that

$$\begin{aligned} AP &= \begin{bmatrix} (3)(-1/2) + (1)(1) & (3)(-1) + (1)(1) \\ (-2)(-1/2) + (0)(1) & (-2)(-1) + (0)(1) \end{bmatrix} \\ &= \begin{bmatrix} -1/2 & -2 \\ 1 & 2 \end{bmatrix} \end{aligned}$$

and thus that

$$\begin{aligned} P^{-1}AP &= \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} -1/2 & -2 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} (2)(-1/2) + (2)(1) & (2)(-2) + (2)(2) \\ (-2)(-1/2) + (-1)(1) & (-2)(-2) + (-1)(2) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \end{aligned}$$

Since $P^{-1}AP$ is a diagonal matrix, the matrix A is diagonalizable.

Task Use the previous computation to determine the eigenvalues of A .

We calculate that

$$\begin{aligned}c_A(\lambda) &= \det(\lambda I - A) \\&= \det(P(\lambda I - P^{-1}AP)P^{-1}) \\&= \det(P) \det(\lambda I - P^{-1}AP) \det(P)^{-1} \\&= \det(\lambda I - P^{-1}AP) \\&= \det \begin{bmatrix} \lambda - 1 & 0 \\ 0 & \lambda - 2 \end{bmatrix} \\&= (\lambda - 1)(\lambda - 2).\end{aligned}$$

Thus $\{1, 2\}$ is the set of eigenvalues of A .

Task Use the previous computation to calculate A^5 .

Let's start by computing smaller powers of A .
We find that

$$\begin{aligned}A^2 &= A \cdot A = (PDP^{-1})(PDP^{-1}) \\&= PD(P^{-1}P)DP^{-1} = PD^2P^{-1}\end{aligned}$$

and

$$\begin{aligned} A^3 &= A \cdot A^2 = (PDP^{-1})(PD^2P^{-1}) \\ &= PD(P^{-1}P)D^2P^{-1} = PD^3P^{-1} \end{aligned}$$

Continuing this calculation, noticing the pattern, we conclude that $A^5 = PD^5P^{-1}$.

Multiplying diagonal matrices is easy, i.e.

$$\begin{bmatrix} a_1 & 0 \\ 0 & b_1 \end{bmatrix} \begin{bmatrix} a_2 & 0 \\ 0 & b_2 \end{bmatrix} \cdots \begin{bmatrix} a_m & 0 \\ 0 & b_m \end{bmatrix} = \begin{bmatrix} a_1 \cdots a_m & 0 \\ 0 & b_1 \cdots b_m \end{bmatrix}.$$

Thus

$$D^5 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^5 = \begin{bmatrix} 1^5 & 0 \\ 0 & 2^5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 32 \end{bmatrix}.$$

We use this computation to compute A^5 , i.e. we calculate that

$$PD^5 = \begin{bmatrix} -1/2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 32 \end{bmatrix} = \begin{bmatrix} -1/2 & -32 \\ 1 & 32 \end{bmatrix}$$

and thus that

$$PD^5P^{-1} = \begin{bmatrix} -1/2 & -32 \\ 1 & 32 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -2 & -1 \end{bmatrix} \\ = \begin{bmatrix} 63 & 31 \\ -62 & -30 \end{bmatrix}.$$

The previous activity shows why diagonalizing a matrix may be useful. But given a matrix A , how do we potentially find P such that $P^{-1}AP$ is a diagonal matrix?

Activity

Suppose $P^{-1}AP = D$ for some diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be the columns of P . Note that the equation $P^{-1}AP = D$ holds if and only if $AP = PD$.

Task What happens when I take PD ?

$$PD = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Matrix multiplication gives us:

$$PD = [\lambda_1 \mathbf{v}_1 \ \lambda_2 \mathbf{v}_2 \ \cdots \ \lambda_n \mathbf{v}_n]$$

So PD is the matrix whose columns are the vectors $\mathbf{v}_i$ scaled by their corresponding λ_i values.

Task Calculate AP and PD in terms of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, the scalars $\lambda_1, \dots, \lambda_n$ and the matrix A .

When we take AP , we compute that

$$AP = A [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] = [A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n].$$

When we calculate PD , we compute that

$$\begin{aligned} PD &= [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n] \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \\ &= [\lambda_1 \mathbf{v}_1 \ \cdots \ \lambda_n \mathbf{v}_n]. \end{aligned}$$

Task So what conditions on P and D are necessary and sufficient for

the equation $AP = PD$ to hold?

For the equation $AP = PD$ to hold, we need $A\mathbf{v}_i = \lambda_i\mathbf{v}_i$ for each i . Thus we need each column of P to be an eigenvector of A , whose eigenvalue is the corresponding entry of the diagonal matrix D .

Theorem An $n \times n$ matrix A is *diagonalizable* if and only if A has n eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ such that the matrix P with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ is invertible. In this case, $P^{-1}AP = D$, where D is a diagonal matrix such that the elements of D on the main diagonal are the *eigenvalues* of A .

Remark Recalling Invertibility Criteria, the matrix P in Theorem thm-7-1-diagonalizability is invertible if and only if the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ form a basis for $\mathbb{R}^n$.

If $\mathbb{R}^n$ has a basis of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with corresponding eigenvalues $\lambda_1, \dots, \lambda_n$, then every vector $\mathbf{v} \in \mathbb{R}^n$ can be written as a linear combination of these eigenvectors, i.e. there exists constants $a_1, \dots, a_n$ such that

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n.$$

Thus

$$\begin{aligned}
T_A(\mathbf{v}) &= A\mathbf{v} \\
&= A(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n) \\
&= a_1A\mathbf{v}_1 + a_2A\mathbf{v}_2 + \cdots + a_nA\mathbf{v}_n \\
&= a_1\lambda_1\mathbf{v}_1 + a_2\lambda_2\mathbf{v}_2 + \cdots + a_n\lambda_n\mathbf{v}_n \\
&= \lambda_1(a_1\mathbf{v}_1) + \lambda_2(a_2\mathbf{v}_2) + \cdots + \lambda_n(a_n\mathbf{v}_n)
\end{aligned}$$

In other words, T_A ‘stretches’ the components of $\mathbf{v}$ along each eigenvector by the corresponding eigenvalue.

Activity

Consider the matrix

$$M = \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix}.$$

Verify that

$$\mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

are eigenvectors of M corresponding to different eigenvalues. What are the eigenvalues?

We compute that

$$\begin{aligned}
 M\mathbf{u} &= \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 5/3 - 2/3 \\ -1/3 + 4/3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 &= 1 \cdot \mathbf{u}
 \end{aligned}$$

and

$$\begin{aligned}
M\mathbf{v} &= \begin{bmatrix} 5/3 & -2/3 \\ -1/3 & 4/3 \end{bmatrix} \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} \\
&= \begin{bmatrix} 5/3 - 2/3 \cdot (-1/2) \\ -1/3 + 4/3 \cdot (-1/2) \end{bmatrix} \\
&= \begin{bmatrix} 5/3 + 1/3 \\ -1/3 - 2/3 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\
&= 2 \cdot \begin{bmatrix} 1 \\ -1/2 \end{bmatrix} \\
&= 2 \cdot \mathbf{v}.
\end{aligned}$$

Therefore, $\mathbf{u}$ is an eigenvector with eigenvalue $\lambda_1 = 1$ and $\mathbf{v}$ is an eigenvector with eigenvalue $\lambda_2 = 2$.

textbook figure fig-7-1-eigenvector-transformation illustrates the effect of the matrix M from Activity ex-eigenvector-verification on a grid in $\mathbb{R}^2$. In particular, it highlights an important geometric interpretation of eigenvectors and eigenvalues; when we apply the linear map defined by the matrix M to any vector in $\mathbb{R}^2$, the matrix map stretches eigenvector directions by their eigenvalues.

Remark: Non-diagonalizable matrices Consider the rotation matrix

$$A_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

where θ is not a multiple of 180° . For any nonzero vector $\mathbf{v} \in \mathbb{R}^2$, the vector $A_\theta \mathbf{v}$ is obtained by rotating $\mathbf{v}$ counter-clockwise by an angle θ , and so $A\mathbf{v}$ is *never* on the line spanned by $\mathbf{v}$, i.e. it is never true that the equation $A\mathbf{v} = \lambda\mathbf{v}$ for any λ , and any non-zero $\mathbf{v}$.

Thus the matrix A_θ has no eigenvectors or eigenvalues. It therefore is a matrix that is not diagonalizable. This makes geometric sense: we cannot find special directions in $\mathbb{R}^2$ so that the action of rotating vectors in the plane is obtained by stretching in those special directions.

Non-diagonalizable matrices illustrates why not every matrix is diagonalizable. A matrix is diagonalizable precisely when it has a basis of eigenvectors, but this is not always possible.

Activity

Let

$$A = \begin{bmatrix} 3 & 0 \\ -1 & -4 \end{bmatrix}.$$

Is A diagonalizable?

We need to check if A has a basis of eigenvectors. In Activity act-7-1-find-eigenvalues, we showed that A has eigenvectors

$$\mathbf{v}_1 = \begin{bmatrix} -7 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The two vectors are linearly independent as they are not parallel, and thus form a basis for $\mathbb{R}^2$. Therefore, A is diagonalizable.

Activity

Let

$$A = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}.$$

Is A diagonalizable?

We start by calculating the characteristic polynomial of A , that

$$\det(\lambda I - A) = \det \begin{bmatrix} \lambda - 4 & -1 \\ 0 & \lambda - 4 \end{bmatrix} = (\lambda - 4)^2.$$

Thus $\lambda = 4$ is the only eigenvalue of A .

To find the eigenvectors of A , we solve the equation $(4I - A)\mathbf{v} = \mathbf{0}$, which we expand as

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The equation has only a single basic solution, i.e.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Recalling Fact fact-7-1-every-eigenvector-basic, every eigenvector of A is a scalar multiple of this vector. So A cannot have a basis of eigenvectors, and so A is not diagonalizable.